2.) Diketahui $F(x) = \frac{1}{4}x^4 + \frac{1}{3}x^3 - x^2 + k$ ↘ $F(x) = 1/4x^4 + 1/3x^3 - x^2 + 10$

a. Persamaan garis singgung kurva F(x) melalui $x = 1$

$F'(x) = x^3 + x^2 - x$

$y - y_1 = m(x - x_1)$

$y - 0 = x^3 + x^2 - x(x-1)$

$y = x^4 - x^3 + x^3 - x^2 - 2x^2 + 2x$

$y = x^4 - 3x^2 + 2x$

b. Tent. Interval Fungsi naik dan Fungsi turun

F. naik → $F'(x) > 0$

$x^3 + x^2 - x > 0$

$x(x^2 + x - 2) > 0$

$x > 0 \; ; \; (x+2)(x-1) > 0$

$x > -2 \cup x > 1$

$x > 0 \; ; \; x > -2 \; ; \; x > 1$

Naik → $-2 \leq x \leq 0$

F. turun → $F'(x) < 0$

$x^3 + x^2 - x < 0$

$x(x^2 + x - 2) < 0$

$x(x+2)(x-1) < 0$

$x < 0 \; ; \; x < -2 \; ; \; x < 1$

Turun → $0 \leq x \leq 1$

c. Titik Stasioner yang diminta $F(x) = 0$

$x^3 + x^2 - x = 0$ $x = 0$

$x(x^2 + x - 2)$ $x = -2$

$x(x+2)(x-1) = 0$ $x = 1$

$F''(x) = 3x^2 + 2x$

$F''(0) = +$ (maksimum)

$F''(-2) = -$ (minimum)

$F''(1) = -$ (minimum)

Nilai maks → 0 pada $x = 0$

$F(0) = 0$ (0,21) maksimum

$F(-2) = -8/3$ (-2, 18,33) minimum

$F(1) = -5/12$ (1, 20,538) minimum

Dipindai dengan CamScanner